

14	D	The equation of the tide can be modelled as $1.5 = -\cos(2\pi t/12.5) + 2$, where 2 is the equilibrium level, with an amplitude of 1 m and period of 12.5 hours. Solving gives $t = 2.08$ or 2.1 hours.
15	C	Max $a = (2\pi \times 120)^2 \times 40 \times 10^{-3} = 22716 \text{ ms}^{-2}$ and max $v = (2\pi \times 120) \times 40 \times 10^{-3} = 30.1 \text{ ms}^{-1}$